How to Read/Summarize/Write a Research Paper

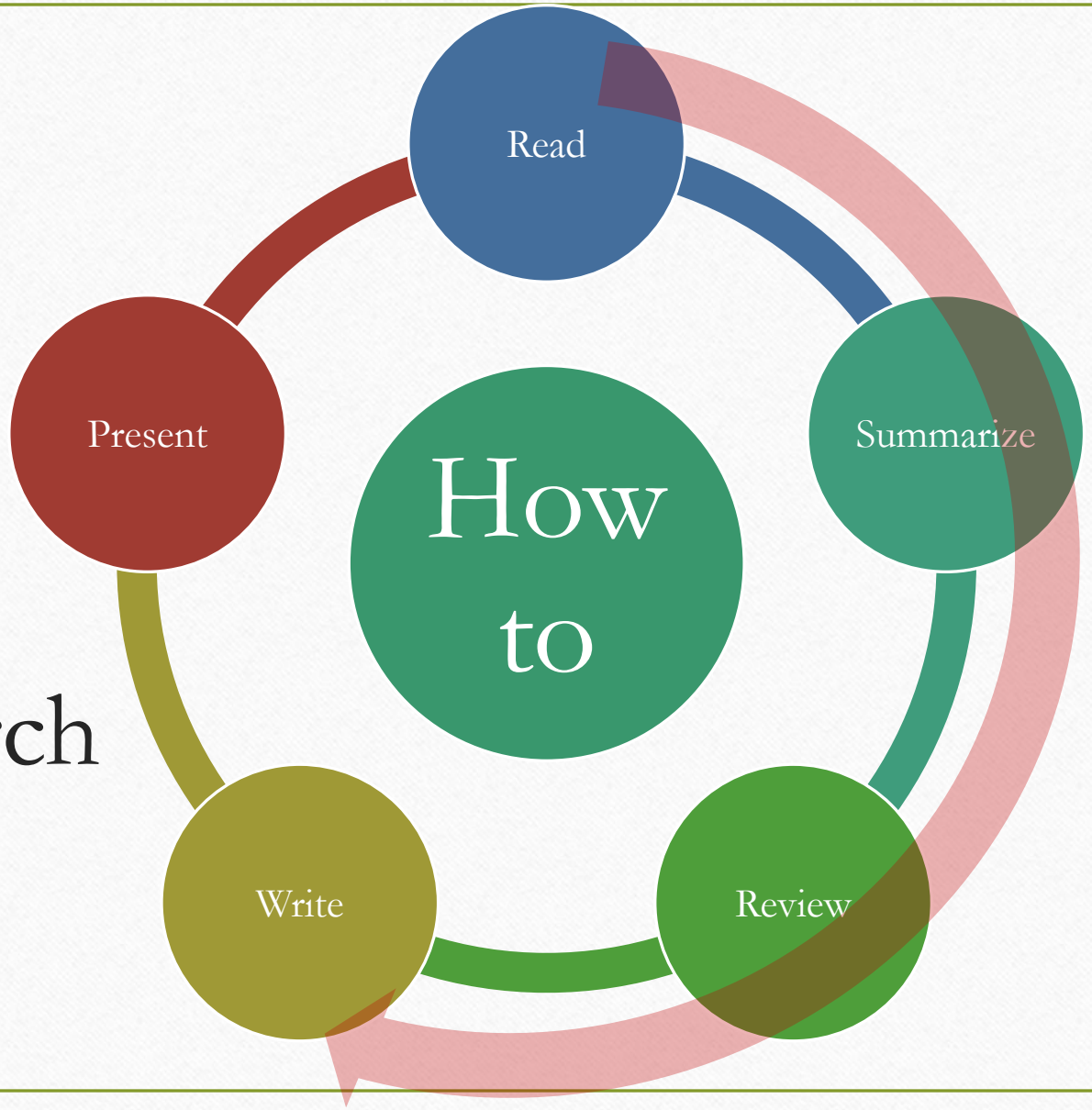
CSE 300: Technical Writing & Presentation

Course Teachers: MSB, MTB, RR, MSRJ, AW

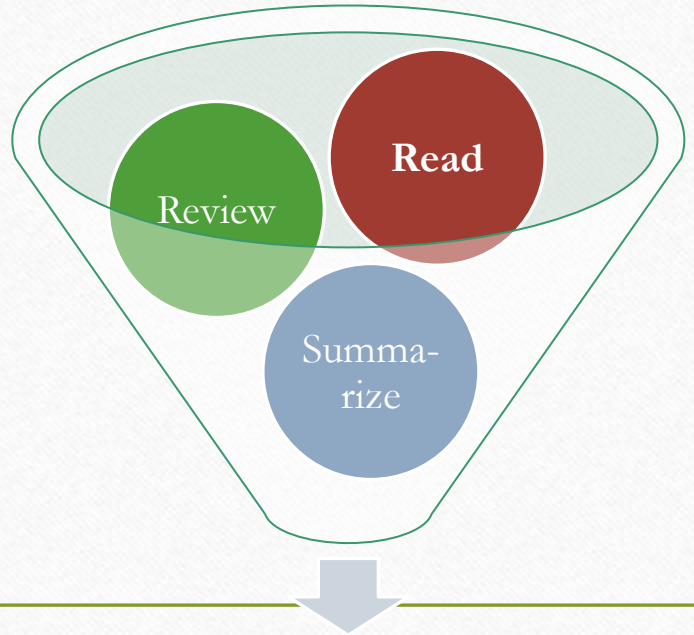
Presenter: MSRJ



How to ___ a Research Paper?

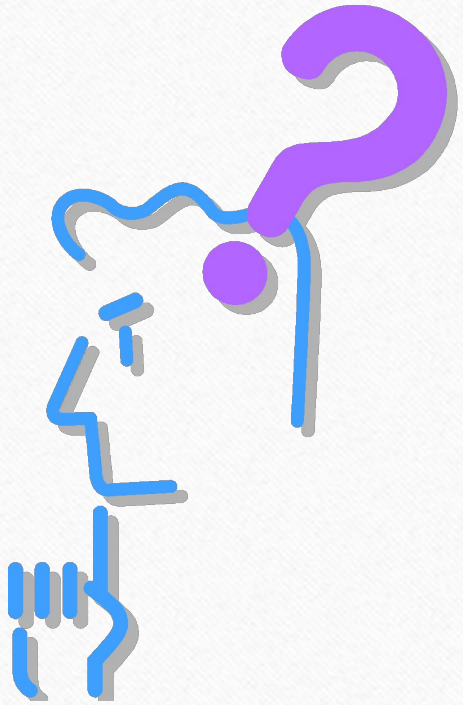


Section 1: READ



How to Read a Research Paper?


WHY
WHAT
HOW



WHY







Review

Conference

Journal

Class

Own

Field

Keep

Up-to-dat

e

Literature

review

New

Field









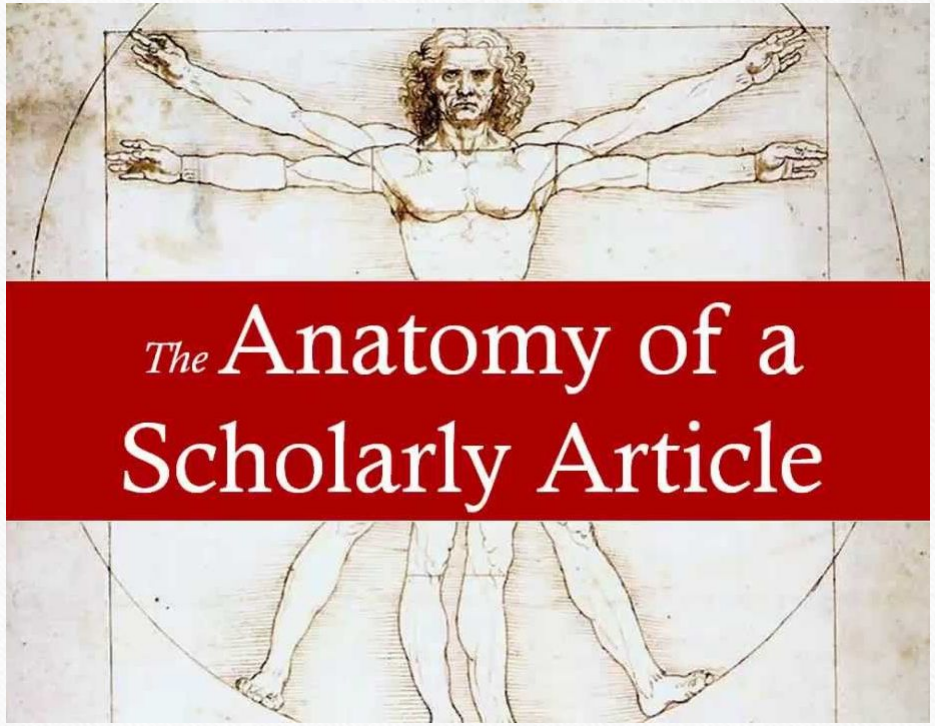
Topic
- Right paper?
- Exp. vs. Theor.?

Venue
- Reputed venue?
- Jour. vs. Conf.?

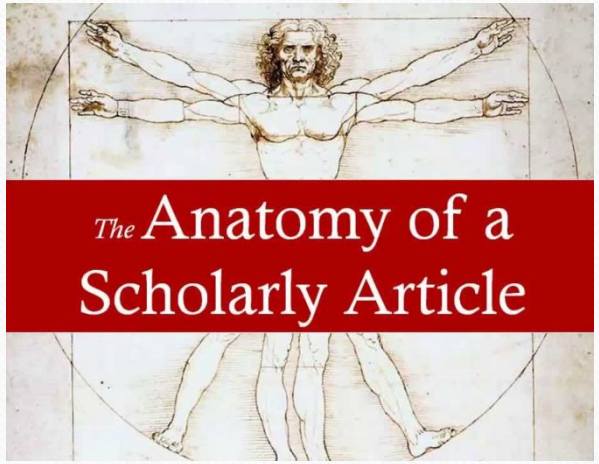
Quality
- Citation?
- Impact?

Authors
- Reputed?
- Recent Works?


HOW


The Anatomy of a
Scholarly Article







ANATOMY OF A SCIENTIFIC ARTICLE

ABSTRACT

A very short, clear, and concise summation of the entire paper.

INTRODUCTION

The introduction reveals the purpose of the paper, starting broadly and narrowing down to a specific thesis or research question.

MATERIALS & METHODS

Can be divided into two separate sections. Clearly outlines how the study/experiment was carried out.

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Objectively presents key results without interpretation in an orderly, logical, sequence. Includes text, graphs, statistics, etc.

DISCUSSION

Interprets results in light of what was already known about the subject, and to explain new understanding of the problem after taking found results under consideration.

CONCLUSION

Reinforces major claims or interpretations. The author will try to indicate the significance of the major claim/interpretation beyond the scope of the paper.

IMRaD





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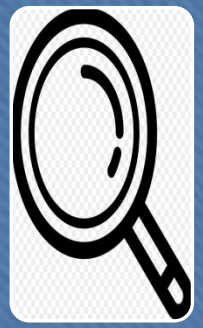
DISCUSSION

Interprets results in light of what was already known about the subject, and to explain new understanding of the problem after taking found results under consideration.

CONCLUSION

Reinforces major claims or interpretations. The author will try to indicate the significance of the major claim/interpretation beyond the scope of the paper.

Multi-Pass Approach



General Idea

- Pass 1

Content

(No Details)

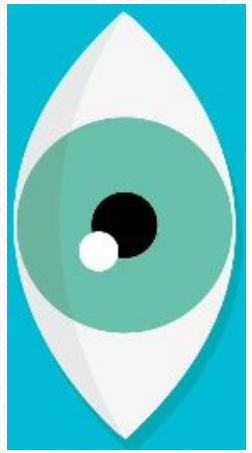
- Pass 2

Depth

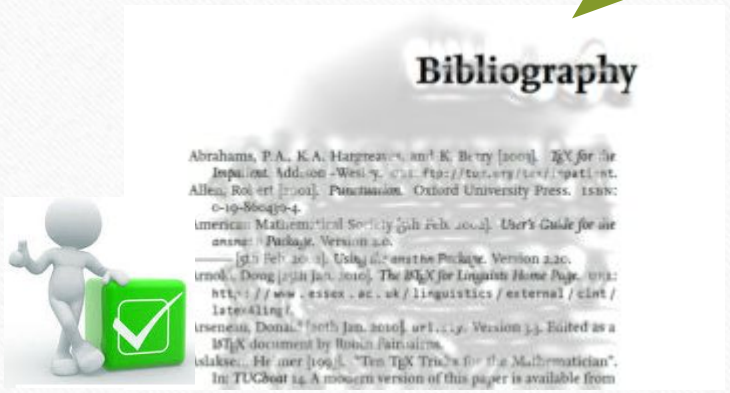
- Pass 3

Pass 1









ANATOMY OF A SCIENTIFIC ARTICLE

ABSTRACT

A very short, cle… the entire paper.

INTRODUCTION

The introductio… aper, starting broadly … cific thesis or research question.

MATERIALS & METHODS

Can be divided into two separate sections. Clearly outlines how the study/experiment was carried out.

RESULTS

Objectively p… key results … interpretation in an or… logical, sequenc… des text, graphs, statisti…, etc.

DISCUSSION

Interprets results in light of what was already known about the subject, and to explain new understanding of the problem after taking found results under consideration.

CONCLUSION

Reinforces major … . The author will try to indicate … ajor claim/interpretati… the paper.

Headings
Sub-heading

Pass 1



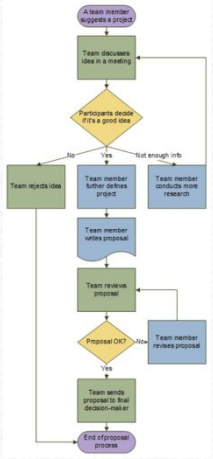


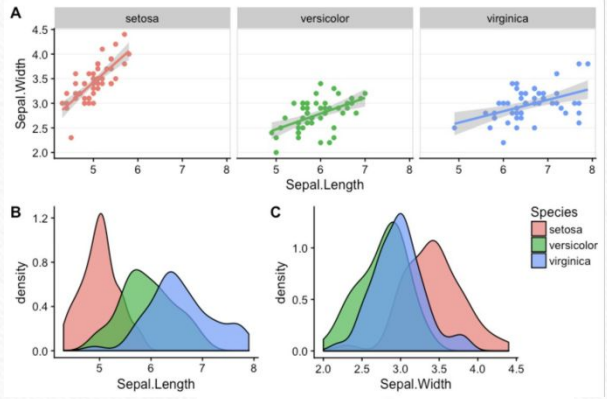


- Context
- Correct (Assumptions)
- Category
- 3Cs
- Contributions
- Clarity

Pass 2









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RES





DISCU

Interprets results in light of what was already known about the subject, and to explain new understanding of the problem after taking found results under consideration.

CONCLUSION

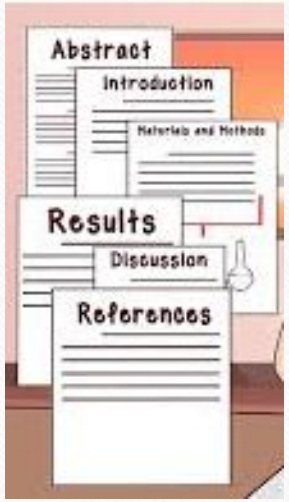
Reinforces major claims or interpretations. The author will try to indicate the significance of the major claim/interpretation beyond the scope of the paper.

Pass 2









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Pass 2

Set the paper aside

Return to the paper later

New

Poor

Reading background material

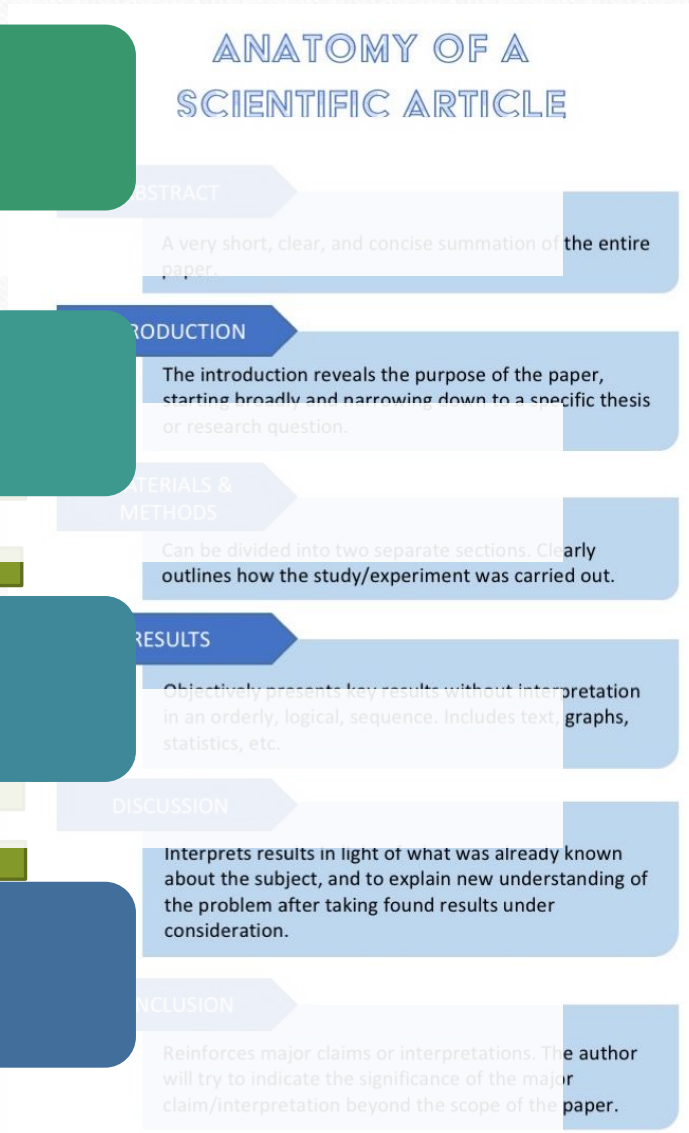
Jargons

Forward ref.



Persevere and go to next pass

Not Understandable



Pass 3









Make author assumptions

Re-create the work

Identify failings and assumptions

Insight

Pinpoint issues

Literature Survey

Pass 3

Literature Survey





Google Scholar

3-5 Papers

Check Related Works Section

Pointer to a Survey paper

Read the Survey

STAGE 1

Pass 3

Literature Survey

- Check citations
- Repeated Authors
- Find web pages

Key Persons

Top venues

- Where the key persons publish
  - Conferences
  - Journals

- Download papers
- Read
  - Multi-pass

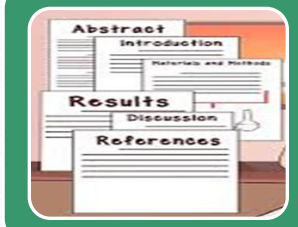
Proceedings/ Archives



Pass 3







Reconstruct the entire structure of the paper from memory.



Identify its strong and weak points.



pinpoint implicit assumptions, missing citations.



Identify potential issues with experimental or analytical techniques.





Read Critically



Read Creatively



Take Notes



Read Critically



















Read Creatively

- Any Good Ideas?
- Idea Extensions?
- Influential results?
- Possible Improvements?



Take Notes



Your will Forget



Electronic Notes



Written Notes

Summarization

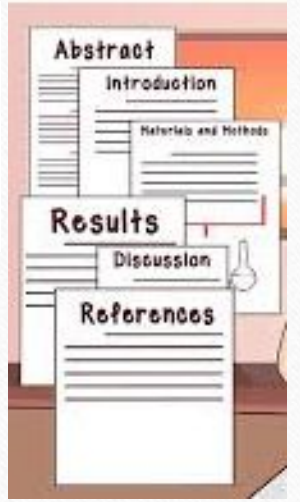










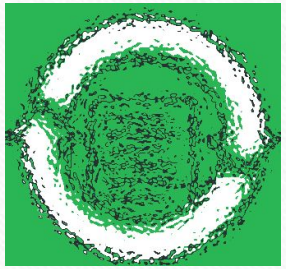


Summarization





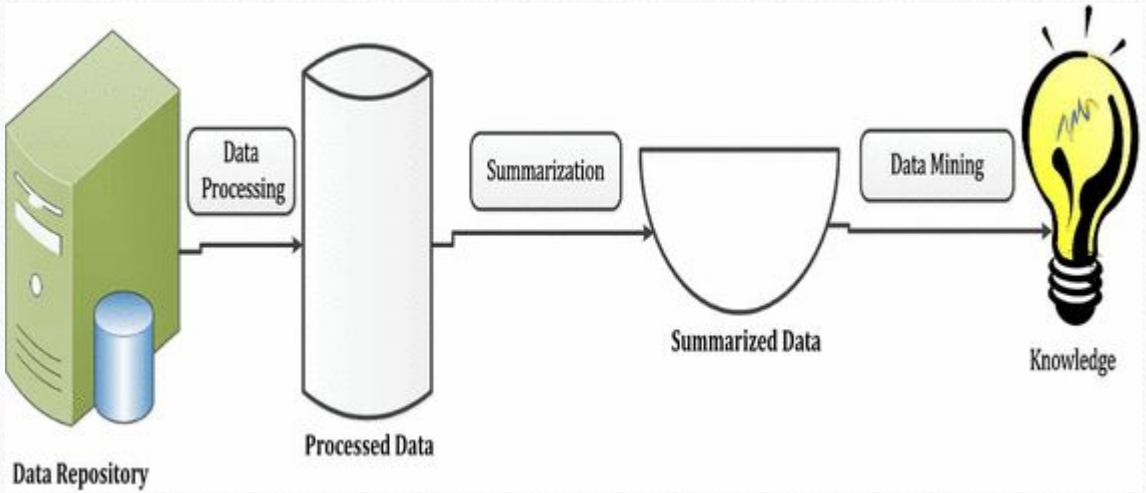






Automatic Summarization

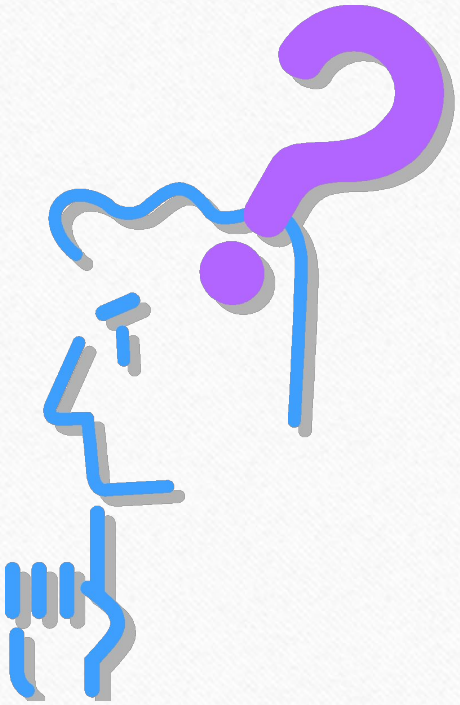




Section 2: WRITE

How to write a Research Paper?

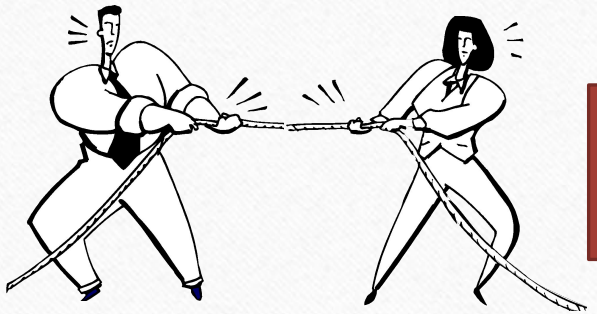

WHAT
WORTH
HOW





What do I have to say in the paper?

Is it one or more papers?



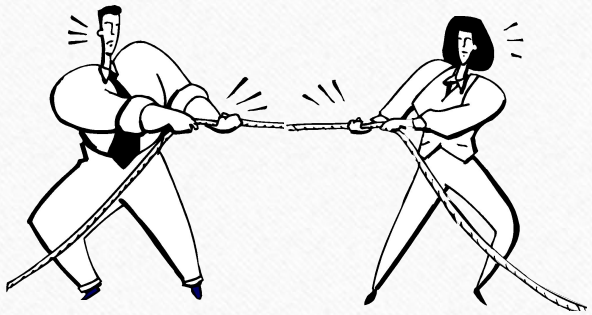
Salami Science???



Putting too much in one paper makes it diffuse and less compelling

Try stating the question and its answer in one simple sentence.

A single question clearly stated with adequate evidence for the answer.



Salami Science

Reporting the results of a single study in 2 or more manuscripts.

- Wastes Journal space
- Time waste for Reviewers
- Time waste for Readers



3 Reasons

- Publish or Perish
- Quantity; Not Quality
- More Support

When Justified?

- Different audience targets
- Too long a manuscript



Is the paper worth writing?

What is in the literature?

It is a lot of work!!!

20-30 drafts!!!

Don't do this unless it is worth.



OK, So you want to/need to write a paper…

So, How to Proceed?





More Questions

Format

Article — Original Research

Short Comm. — Letter

Review — Like a focused mini book

Journal

Conference

Which J/C

EASY

Who cares?

Venue

Audience

All related to each other



Venue Scope?

Allowed Format?

Audience Match?

Impact?

Take advice

Mentor

senior researchers

Consequence of Wrong Decision

time lost

failure to publish



Choice Parameters



Impact Factor

SCImago Rank

Editorial Board



https://beallslist.net/

Indexing

…



Research article as a critical Argument

Research Article



Problem Statement posing a question



Presentation of evidence



Assessment of the validity of the evidence in the face of ..

- strengths/weaknesses
- other evidence

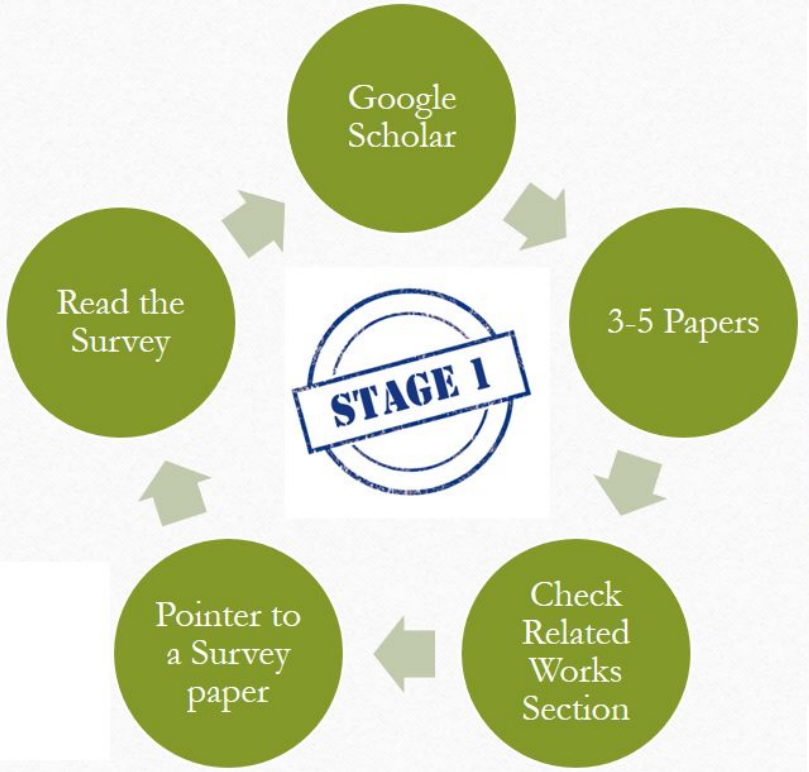


Conclusions



Literature Survey

Recall…



- Check citations
- Repeated Authors
- Find web pages

Key Persons

Top venues

- Where the key persons publish
  - Conferences
  - Journals

- Download papers
- Read
  - Multi-pass

Proceedings/ Archives

STAGE 2



Literature Survey

What has been done and what can you say that's new?

Be thorough in your search:---a high sensitivity/low specificity search.

- Low False -ve
- Misses only a few

High Sens.



- High False +ve
- Too eager to find +ve

Low Spec.





Title Page

Fix earlier than later

Order

- Alphabetic
- Corr.
- Senior
- First

Avoid hurt feeling

Assign Responsibilities

Title

Tentative

Author Order



Sources of Support

Acknowledgement



Create an outline first

Plan on multiple drafts

- Filename with dates
- One filename written over with new draft

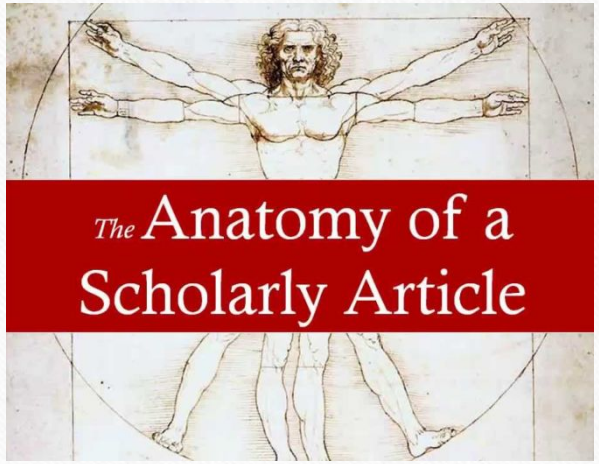




Tables/figures early

- prompt more analysis

Deadlines for you and coauthors





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Introduction

5Qs (CS)

What is the problem

Why is it interesting and important?

Why is it hard? (E.g., why do naive approaches fail?)

Why hasn't it been solved before? (Or, what's wrong with previous proposed solutions? How does mine differ?)

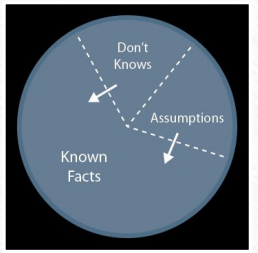
What are the key components of my approach and results?



Draw audience in; be provocative

Target journal specific audience





Identify gaps in knowledge



End with question/hypothesis





Methods

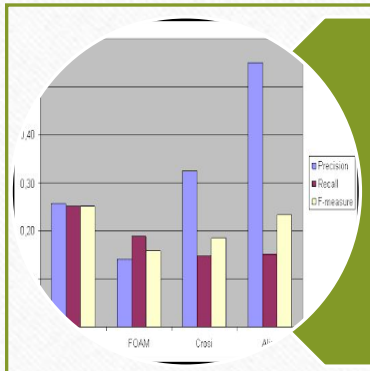
Should include subheadings

Should be comprehensible by nontechnical reader

Should be replicable

Write clearly enough

Past tense



Results

Fig. vs. Tab.

Mix-up

Suppl.

- Organize around tables/figures
-  Past tense
- No interpretation; just the facts!
- Tables/Figures should be self-contained



Discussions

Could constitute a separate section

1st part
- Answer main question/ hypothesis
- Interpret the main results
- Can take help from the Results section content

2nd, 3rd, …
- Evidence pros & cons
- Implications of findings (be conservative)
- Other findings of your study
- Limitations of your study

Last part
- Concluding remarks
- Future directions



Writing Tips

Tense



When quoting *previously published work*, refer to it in *present tense.*

When describing your *own study*, refer to work in *past tense*

Methods and Results in Past tense



Writing Tips

Voice



Some branches dictate passive voice only

Some allow (sometimes prefer) active voice

Best practice: Mixed voice



Writing Tips

Brevity	Be brief. Make every word count.
	If a word does not support your point, cut it out.
	Use shorter and more direct phrases wherever possible.



Writing Tips

Section

Paragraph

Sentence

Word

For each section in the paper

Ask what role it serves

Whether it contributes to the paper's main point

If not, delete it



Writing Tips

Section | Paragraph | Sentence | Word

For each paragraph in a section

Does it have a single point

Whether it contributes to the goal of the section.

If not, rewrite or delete it.



Writing Tips

Section → Paragraph → Sentence → Word

For each sentence in a para

- Does it have a single point
- Whether it strengthens the para
- If not, rewrite or delete it



Writing Tips

Section → Paragraph → Sentence → Word

Whether it strengthens the point

For each word in a sentence

If not, rewrite or delete it



Proofread

Are terms used consistently throughout?

Do numbers in abstract match numbers in text and tables?

Do citations in text match references?

Are Syntax and Grammar acceptable



Abstract

Like a paper

- requires many drafts
- most oriented to presenting argument concisely

Mini-paper

Introduction (usually 1-2 sentences)

Methods (often longest part)

Results (no tables/figures)

Discussion is limited to concluding statement

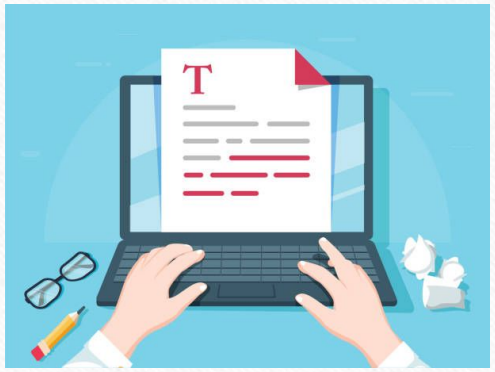
Section 3: Seven Suggestions

Writing a CS Paper

Seven simple, actionable suggestions
that will make your papers better.

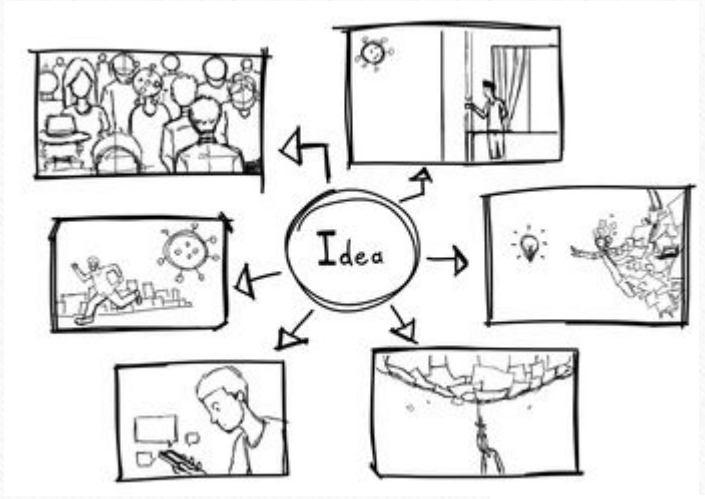
Courtesy: Simon Peyton Jones

Microsoft Research Cambridge


T


idea

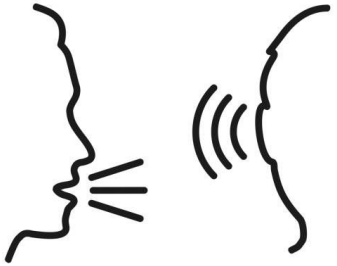



Idea


CONTRIBUTION


EARLY
LATE





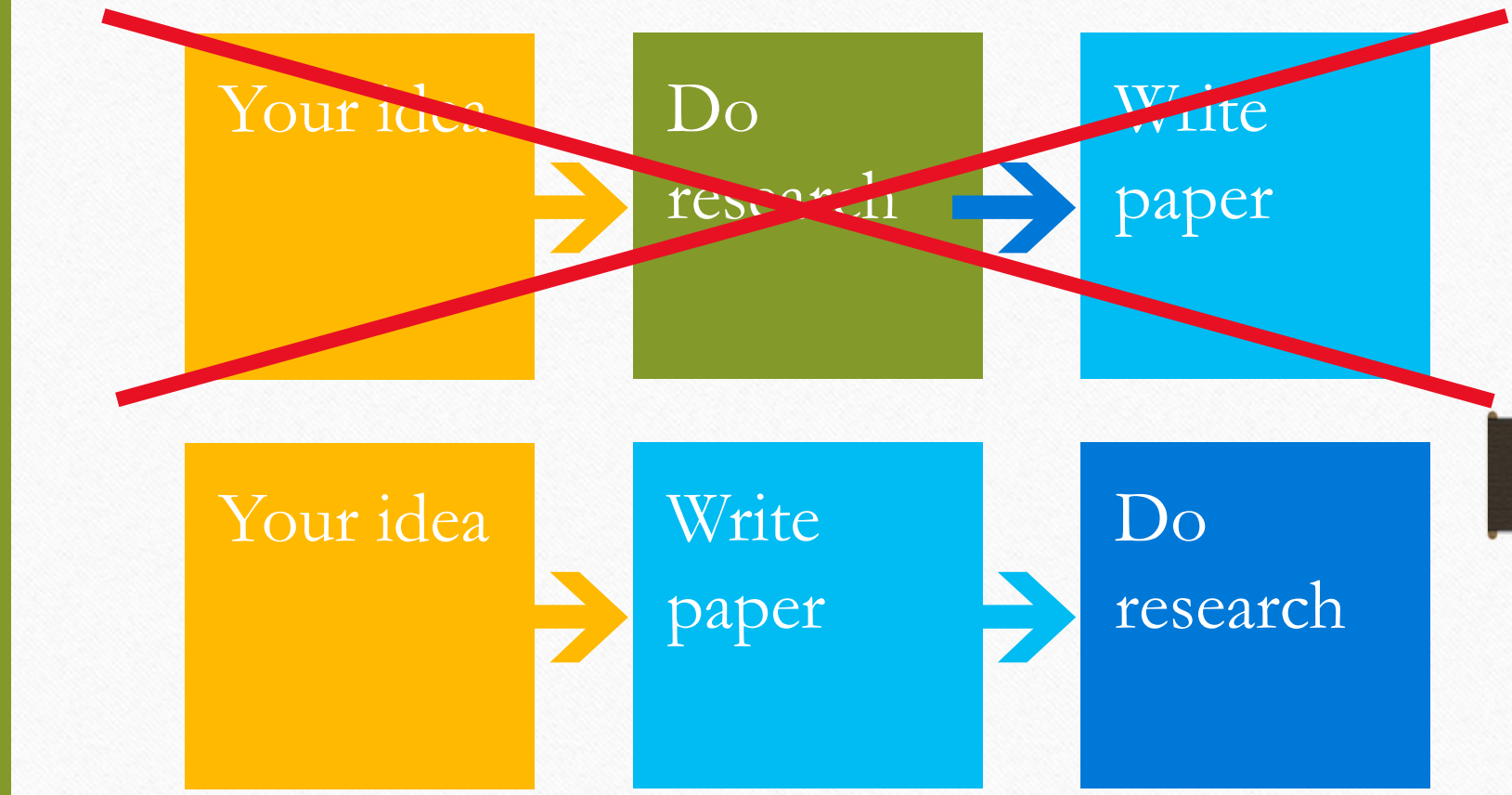


Don’t wait: write

Writing papers: model 1



Writing papers: model 2



Writing papers: model 2



- Forces us to be clear, focused
- Crystallises what we don't understand
- Opens the way to dialogue with others: reality check, critique, and collaboration

Writing papers: model 2



Writing papers is a primary mechanism for doing research (not just for reporting it)



Identify your key idea

Your goal: to convey a useful and re-usable idea

- You want to infect the mind of your reader with your idea, like a virus

The greatest ideas are (literally) worthless if you keep them to yourself

Do not be intimidated

Fallacy

You need to have a fantastic idea before you can write a paper. (Everyone else seems to.)

Write a paper, and give a talk, about any idea, no matter how weedy and insignificant it may seem to you

Do not be intimidated

- Writing the paper is how you develop the idea in the first place
- It usually turns out to be more interesting and challenging that it seemed at first

Write a paper, and give a talk, about any idea, no matter how weedy and insignificant it may seem to you

The idea

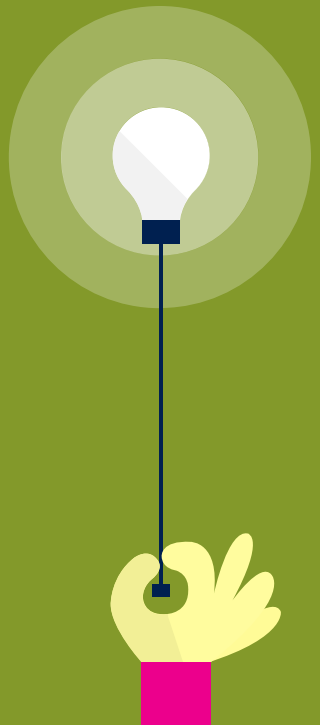
Idea:

A re-usable insight,

useful to the reader

- Your paper should have just one "ping": one clear, sharp idea
- You may not know exactly what the ping is when you start writing; but you must know when you finish
- If you have lots of ideas, write lots of papers

Can you hear the “ping”?



- Many papers contain good ideas, but do not distil what they are.
- Make certain that the reader is in no doubt what the idea is. Be 100% explicit:
  - “The main idea of this paper is....”
  - “In this section we present the main contributions of the paper.”



Tell a Story

Your narrative flow

Imagine you are explaining at a whiteboard

- Here is a problem
- It's an interesting problem
- It's an unsolved problem
- Here is my idea
- My idea works (details, data)
- Here's how my idea compares to other people's approaches

Structure (conference paper)

- Title (1000 readers)
- Abstract (4 sentences, 100 readers)
- Introduction (1 page, 100 readers)
- The problem (1 page, 10 readers)
- My idea (2 pages, 10 readers)
- The details (5 pages, 3 readers)
- Related work (1-2 pages, 10 readers)
- Conclusions and further work (0.5 pages)

Contrast with the IMRaD format….



Nail your contributions to the mast

The introduction (1 page)

- Describe the problem
- State your contributions

...and that is all

RECALL

5Qs	*What is the problem*
(CS)	*Why is it interesting and important?*
	Why is it hard?
	Why hasn't it been solved before?
	What are the key components of my approach and results?

ONE PAGE!

Describe the problem

1 Introduction

There are two basic ways to implement function application in a higher-order language, when the function is unknown: the *push/enter* model or the *eval/apply* model [11]. To illustrate the difference, consider the higher-order function **zipWith**, which zips together two lists, using a function **k** to combine corresponding list elements:

```
zipWith :: (a->b->c) -> [a] -> [b] -> [c]
zipWith k []     []     = []
zipWith k (x:xs) (y:ys) = k x y : zipWith xs ys
```

Here **k** is an *unknown function*, passed as an argument; global flow analysis aside, the compiler does not know what function **k** is bound to. How should the compiler deal with the call **k x y** in the body of **zipWith**? It can't blithely apply **k** to two arguments, because **k** might in reality take just one argument and compute for a while before returning a function that consumes the next argument; or **k** might take three arguments, so that the result of the **zipWith** is a list of functions.

Use an example to introduce the problem

State your contributions

- Write the list of contributions first
- The list of contributions drives the entire paper: the paper substantiates the claims you have made
- Reader thinks "gosh, if they can really deliver this, that's be exciting; I'd better read on"

State your contributions

Do not leave the reader to guess what your contributions are!

Which of the two is best in practice? The trouble is that the evaluation model has a pervasive effect on the implementation, so it is too much work to implement both and pick the best. Historically, compilers for strict languages (using call-by-value) have tended to use eval/apply, while those for lazy languages (using call-by-need) have often used push/enter, but this is 90% historical accident — either approach will work in both settings. In practice, implementors choose one of the two approaches based on a qualitative assessment of the trade-offs. In this paper we put the choice on a firmer basis:

- We explain precisely what the two models are, in a common notational framework (Section 4). Surprisingly, this has not been done before.
- The choice of evaluation model affects many other design choices in subtle but pervasive ways. We identify and discuss these effects in Sections 5 and 6, and contrast them in Section 7. There are lots of nitty-gritty details here, for which we make no apology — they were far from obvious to us, and articulating these details is one of our main contributions.

In terms of its impact on compiler and run-time system complexity, eval/apply seems decisively superior, principally because push/enter requires a stack like no other: stack-walking

Bulleted list of contributions

Contributions should be refutable

No!	Yes!
We describe the WizWoz system. It is really cool.	We give the syntax and semantics of a language that supports concurrent processes (Section 3). Its innovative features are...
We study its properties	We prove that the type system is sound, and that type checking is decidable (Section 4)
We have used WizWoz in practice	We have built a GUI toolkit in WizWoz, and used it to implement a text editor (Section 5). The result is half the length of the Java version.

Evidence



- Your introduction makes claims
- The body of the paper provides evidence to support each claim
- Check each claim in the introduction, identify the evidence, and forward-reference it from the claim
- "Evidence" can be: analysis and comparison, theorems, measurements, case studies

No "rest of this paper is..."

- Not:

  "The rest of this paper is structured as follows. Section 2 introduces the problem. Section 3 ...Finally, Section 8 concludes".

- Instead, use forward references from the narrative in the introduction. The introduction (including the contributions) should survey the whole paper, and therefore forward reference every important part.



Related work: later



Structure

- Abstract (4 sentences)
- Introduction (1 page)
- Related work
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Conclusions and further work (0.5 pages)

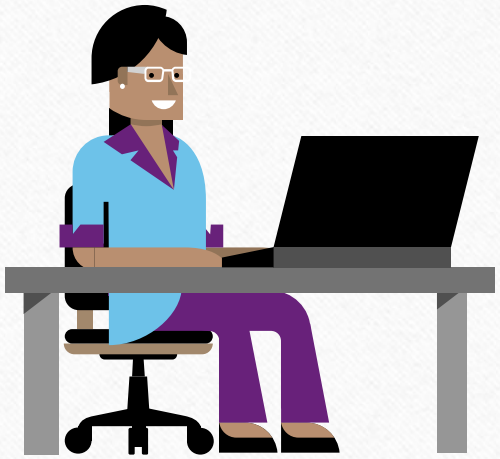
NO!

Structure

- Abstract (4 sentences)
- Introduction (1 page)
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Related work (1-2 pages)
- Conclusions and further work (0.5 pages)

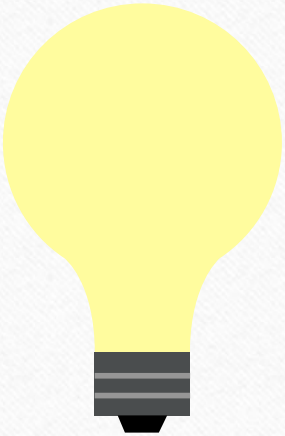
YES!

No related work yet!



Your reader

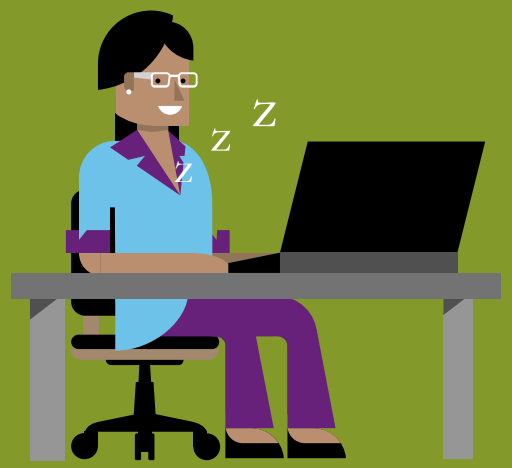
Related work



Your idea

We adopt the notion of transaction from Brown [1], as modified for distributed systems by White [2], using the four-phase interpolation algorithm of Green [3]. Our work differs from White in our advanced revocation protocol, which deals with the case of priority inversion as described by Yellow [4].

No related work yet!



- Problem 1: the reader knows nothing about the problem yet; so your (highly compressed) description of various technical tradeoffs is absolutely incomprehensible
- Problem 2: describing alternative approaches gets between the reader and your idea

Credit

Fallacy

To make my work look good, I have to make other people's work look bad.

The truth: credit is not like money

- Warmly acknowledge people who have helped you
- Be generous to the competition.

  "In his inspiring paper [Foo98] Foogle shows.... We develop his foundation in the following ways..."
- Acknowledge weaknesses in your approach

Giving credit to others does not diminish the credit you get from your paper



Put your readers first

Structure

- Abstract (4 sentences)
- Introduction (1 page)
- The problem (1 page)
- My idea (2 pages)
- The details (5 pages)
- Related work (1-2 pages)
- Conclusions and further work (0.5 pages)

Structure

3. The idea

Consider a bifircuated semi-lattice D, over a $_{j}$ $_{i}$ hyper-modulated signature S. Suppose pi is an element of D. Then we know for every such pi there is an epi-modulus j, such that $p < p$.

- Sounds impressive, but
- Sends readers to sleep, and/or makes them feel stupid

Presenting the idea

- Explain it as if you were speaking to someone using a whiteboard
- Conveying the intuition is primary, not secondary
- Once your reader has the intuition, she can follow the details (but not vice versa)
- Even if she skips the details, she still takes away something valuable

Conveying the intuition

Introduce the problem, and your idea, using EXAMPLES and only then present the general case

- Remember: explain it as if you were speaking to someone using a whiteboard

Using examples

The Simon PJ question: is there any typewriter font?

2 Background

To set the scene for this paper, we begin with a brief overview of the *Scrap your boilerplate* approach to generic programming. Suppose that we want to write a function that computes the size of an arbitrary data structure. The basic algorithm is "for each node, add the sizes of the children, and add 1 for the node itself". Here is the entire code for `gsize`:

```
gsize :: Data a => a -> Int
gsize t = 1 + sum (gmapQ gsize t)
```

The type for `gsize` says that it works over any type `a`, provided `a` is a *data* type — that is, that it is an instance of the class `Data`[1] The definition of `gsize` refers to the operation `gmapQ`, which is a method of the `Data` class:

```
class Typeable a => Data a where
  ...other methods of class Data...
  gmapQ :: (forall b. Data b => b -> r) -> a -> [r]
```

Example right away

Putting the reader first

- Do not recapitulate your personal journey of discovery. This route may be soaked with your blood, but that is not interesting to the reader.
- Instead, choose the most direct route to the idea.



Listen to your readers

Getting help

- Experts are good
- Non-experts are also very good
- Each reader can only read your paper for the first time once! So use them carefully
- Explain carefully what you want ("I got lost here" is much more important than "Jarva is mis-spelt".)

Get your paper read by as many friendly guinea pigs as possible

Getting expert help

- A good plan: when you think you are done, send the draft to the competition saying "could you help me ensure that I describe your work fairly?".
- Often they will respond with helpful critique (they are interested in the area)
- They are likely to be your referees anyway, so getting their comments or criticism up front is Jolly Good.

Listening to your reviewers

Treat every review like gold dust

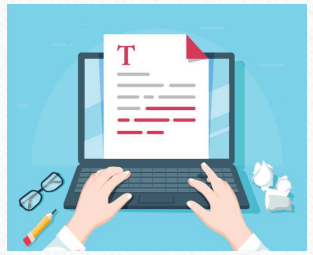
Be (truly) grateful for criticism as well as praise

This is really, really, really hard

But it's really, really, really, really, really, really, really, really, really, really, really important

Listening to your reviewers

- Read every criticism as a positive suggestion for something you could explain more clearly
- DO NOT respond "you stupid person, I meant X".
- INSTEAD: fix the paper so that X is apparent even to the stupidest reader.
- Thank them warmly. They have given up their time for you.

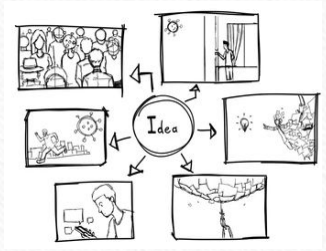




1. Don't wait: write

2. Identify your key idea

Seven simple, actionable suggestions





3. Tell a story

4. Nail your contributions

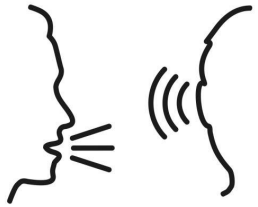




5. Related work: later

6. Put your readers first (examples)





7. Listen to your readers



Section 3: Language and Style

(not comprehensive)

Basic stuff

- Submit by the deadline
- Keep to the length restrictions
  - Do not narrow the margins
  - Do not use 6pt font
  - On occasion, supply supporting evidence (e.g. experimental data, or a written-out proof) in an appendix
- Always use a spell checker

Visual structure

- Give strong visual structure to your paper using
  - sections and sub-sections
  - bullets
  - italics
  - laid-out code
- Find out how to draw pictures, and use them

Visual structure

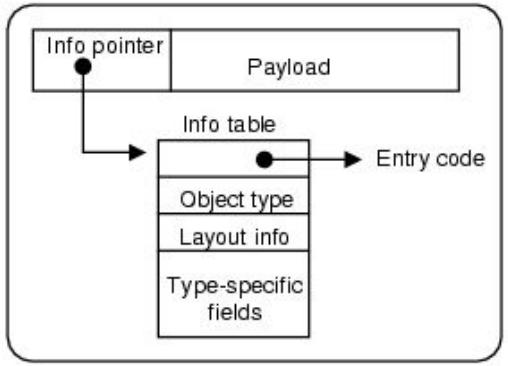


Figure 3. A heap object

The three cases above do not exhaust the possible forms of f. It might also be a *THUNK*, but we have already dealt with that case (rule THUNK). It might be a *CON*, in which case there cannot be any pending arguments on the stack, and rules UPDATE or RET apply.

4.3 The eval/apply model

The last block of Figure 2 shows how the eval/apply model deals with function application. The first three rules all deal with the case of a *FUN* applied to some arguments:

- If there are exactly the right number of arguments, we behave exactly like rule KNOWNCALL, by tail-calling the function. Rule EXACT is still necessary — and indeed has a direct counterpart in the implementation — because the function might not be statically known.
- If there are too many arguments, rule CALLK pushes a *call*

remainder of the object is called the *payload*, and may consist of a mixture of pointers and non-pointers. For example, the object $CON(C\ a_1 \ldots a_n)$ would be represented by an object whose info pointer represented the constructor C and whose payload is the arguments $a_1 \ldots a_n$.

The info table contains:

- Executable code for the object. For example, a *FUN* object has code for the function body.
- An object-type field, which distinguishes the various kinds of objects (*FUN*, *PAP*, *CON* etc) from each other.
- Layout information for garbage collection purposes, which describes the size and layout of the payload. By "layout" we mean which fields contain pointers and which contain non-pointers, information that is essential for accurate garbage collection.
- Type-specific information, which varies depending on the object type. For example, a *FUN* object contains its arity; a *CON* object contains its constructor tag, a small integer that distinguishes the different constructors of a data type; and so on.

In the case of a PAP, the size of the object is not fixed by its info table; instead, its size is stored in the object itself. The layout of its fields (e.g. which are pointers) is described by the (initial segment of) an argument-descriptor field in the info table of the FUN object which is always the first field of a PAP. The other kinds of heap object all have a size that is statically fixed by their info table.

A very common operation is to jump to the entry code for the object, so GHC uses a slightly-optimised version of the representation in Figure 3. GHC places the info table at the addresses *immediately*

Use the active voice

The passive voice is "respectable" but it deadens your paper. Avoid it at all costs.

No!	Yes!
It can be seen that...	We can see that...
34 tests were run	We ran 34 tests
These properties were thought desirable	We wanted to retain these properties
It might be thought that this would be a type error	You might think this would be a type error

Use simple, direct language

No!	Yes!
The object under study was displaced horizontally	The ball moved sideways
On an annual basis	Yearly
Endeavour to ascertain	Find out
It could be considered that the speed of storage reclamation left something to be desired	The garbage collector was really slow


Q&A